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Oefeningenreeks 1C nr. 27.3

Gegeven is de macht $z = \left(3 \operatorname{cis} \frac{\pi}{6}\right)^3$.

$$z = 3^3 \operatorname{cis} \left(3 \cdot \frac{\pi}{6}\right)$$

$$z = 27 \operatorname{cis} \left(\frac{3\pi}{6}\right)$$

$$z = 27 \operatorname{cis} \frac{\pi}{2}$$

$$z = 27 \left(\cos \frac{\pi}{2} + i \sin \frac{\pi}{2}\right)$$

$$z = 27(0 + 1i)$$

$$z = 27i$$

Antwoord: $z = 27i$

References